

Questions			Answers	Marks
6	(a)		Electric field strength at a point is defined by the <u>force per unit charge</u> . Not acceptable Force on a unit charge. [Unit is N] whereas for force per unit charge, the unit is N C^{-1}	B1
			acting on a <u>small positive stationary</u> test charge at that point.	B1
	(b)	(i)	electric field strength inside parallel plate = $\frac{\Delta V}{\Delta x} = \frac{1500-0}{0.015} = 1.0 \times 10^5 \text{ V m}^{-1}$	M1
			Since charge is stationary electric force balances the weight so $qE = mg$ or $N \times 1.6 \times 10^{-19} \times 1.0 \times 10^5 = 4.90 \times 10^{-15} \times 9.81$	M1
			$N = 3$ (given as <u>integer</u>)	A1
		(ii)	<u>initial electric force have the same magnitude as weight.</u>	M1

			Now the direction is same as the weight. So net force on charge is twice the weight.	
			$2g$	A1
			downwards	B1

7	(a)	Faraday's law of electromagnetic induction states that the e.m.f. induced in a
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